**Online Hotel Reservation System**

***A Project Presentation presented to the College of Information Technology Education***
***PHINMA - University of Iloilo***

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***(ITE 298 Information Management)***

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**Three Parts Of a System Overview**

**1. Project Purpose**

The purpose of the **Online Hotel Reservation System** is to provide customers with a convenient and efficient way to book hotel rooms online. The system addresses common problems associated with manual reservations, such as time-consuming booking processes, difficulty checking room availability, and errors in managing reservation records.

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**2. Architecture and Technology Stack**

The system will be developed using the following technologies:

- PHP
- HTML
- CSS
- MySQL

- XAMPP
- Visual Studio Code

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**3. Key Modules and Data Flow**

## 3.1 User Management Module

**Functions:**

- Allows customers to register and log in.
- Stores customer account information securely.
- Verifies user login credentials.

**Data Flow:** Customer enters information → PHP validates the input → MySQL stores or retrieves account information → System displays the result.

## 3.2 Room Management Module

**Functions:**

- Displays available hotel rooms, descriptions, and prices.
- Allows administrators to add, edit, or remove room information.
- Updates room details in the database.

**Data Flow:** Administrator manages room information → PHP processes the changes → MySQL updates room records → Website displays the updated information.

## 3.3 Reservation Module

**Functions:**

- Allows customers to select rooms and enter check-in and check-out dates.
- Checks room availability for the selected dates.
- Calculates the total reservation cost.
- Saves reservation details.

**Data Flow:** Customer selects a room and dates → PHP checks availability in MySQL → System calculates the total cost → Reservation is saved → Booking confirmation is displayed.

## 3.4 Booking Management Module

**Functions:**

- Allows administrators to view customer reservations.
- Enables administrators to confirm or cancel bookings.
- Allows customers to check their reservation status.

**Data Flow:** Administrator views or updates a booking → PHP processes the request → MySQL updates the reservation status → Customer views the latest booking status.

## 3.5 Database Management Module

**Functions:**

- Stores customer accounts, room details, and reservation records.
- Maintains relationships between users, rooms, and reservations.
- Supports accurate retrieval and updating of booking information.

**Data Flow:** System sends a database query → MySQL processes the query → Results are returned to PHP → Website displays the information.

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### 4. Project Summary

The **Online Hotel Reservation System** combines a user-friendly web interface, server-side processing, and database management to simplify hotel reservations. Its five main modules—User Management, Room Management, Reservation, Booking Management, and Database Management—work together to provide customers with a seamless online booking experience while helping hotel administrators efficiently manage hotel operations.

**Two Audiences,Two Vocabularies**

# Reservation Module Feature

## Technical Audience (Developers, DBAs, QA)

**The Reservation Module enables customers to book hotel rooms by selecting an available room and specifying check-in and check-out dates. The system uses PHP to validate user inputs and process reservation requests. MySQL queries are executed to check room availability, calculate booking costs, and store reservation records. The module performs CRUD operations on reservation data and updates room status automatically to prevent double bookings.**

**Data Flow: Customer submits reservation request → PHP validates input → MySQL checks room availability → Total cost is calculated → Reservation record is inserted into the database → Booking status is returned to the user.**

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## Non-Technical Audience (Customers, Managers, Users)

**The Reservation Module allows customers to easily reserve a hotel room online anytime and anywhere. Users can view available rooms, choose their preferred dates, and instantly receive booking confirmation. The system helps reduce waiting time, prevents booking mistakes, and provides a convenient and reliable reservation experience for all customers.**

**Benefits:**

- **Easy online room booking**
- **Fast reservation process**
- **Accurate room availability information**
- **Secure storage of booking records**
- **Convenient access for all users**

**Structuring A Strong Demo**

# Online Hotel Reservation System Demo

## 1. Introduction

**Good day everyone. Our project is an Online Hotel Reservation System designed to provide a convenient and efficient way for people to reserve hotel rooms online. Instead of visiting a hotel or making reservations manually, customers can easily check room availability, select their preferred dates, and complete their bookings through the website.**

**The system helps reduce booking errors, saves time, and improves the overall reservation experience for both customers and hotel administrators.**

## 2. Feature: Reservation Module

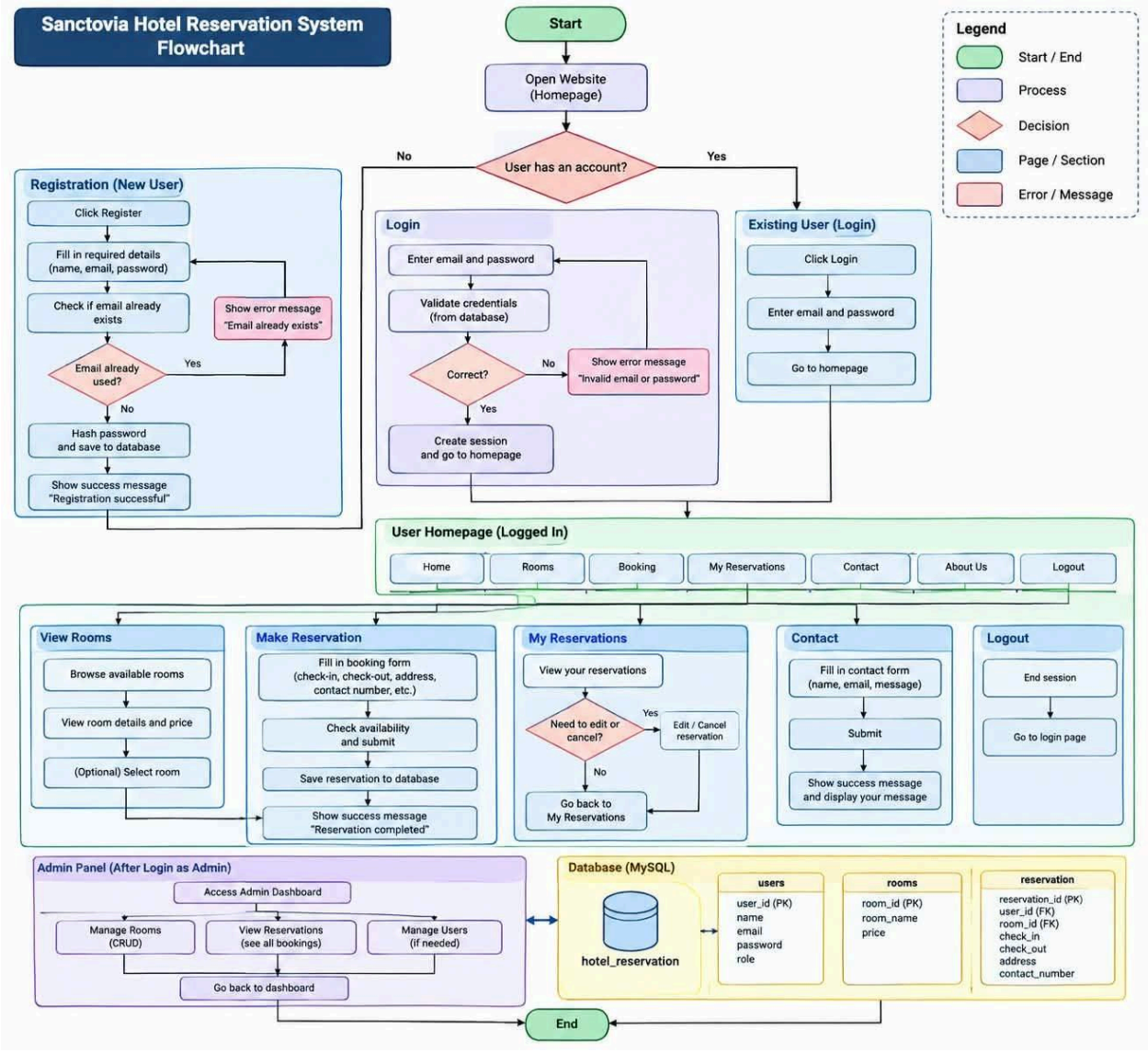
**The main feature we will demonstrate is the Reservation Module.**

**This feature allows customers to:**

- **View available hotel rooms.**
- **Select a preferred room.**
- **Choose check-in and check-out dates.**
- **Make reservations online.**
- **Receive booking confirmation and status updates.**

**The Reservation Module is designed to make hotel booking simple, fast, and accessible for all users. It ensures that customers can reserve rooms with ease while helping the hotel manage reservations more efficiently.**

# 3. Data Flow


Sanctovia Hotel Reservation System Flowchart
Start
Open Website (Homepage)
User has an account?
No
Yes
Legend
Start / End
Process
Decision
Page / Section
Error / Message
Registration (New User)
Click Register
Fill in required details (name, email, password)
Check if email already exists
Email already used?
Yes
No
Show error message "Email already exists"
Hash password and save to database
Show success message "Registration successful"
Login
Enter email and password
Validate credentials (from database)
Correct?
No
Yes
Show error message "Invalid email or password"
Create session and go to homepage
Existing User (Login)
Click Login
Enter email and password
Go to homepage
User Homepage (Logged In)
Home
Rooms
Booking
My Reservations
Contact
About Us
Logout
View Rooms
Browse available rooms
View room details and price
(Optional) Select room
Make Reservation
Fill in booking form (check-in, check-out, address, contact number, etc.)
Check availability and submit
Save reservation to database
Show success message "Reservation completed"
My Reservations
View your reservations
Need to edit or cancel?
Yes
No
Edit / Cancel reservation
Go back to My Reservations
Contact
Fill in contact form (name, email, message)
Submit
Show success message and display your message
Logout
End session
Go to login page
Admin Panel (After Login as Admin)
Access Admin Dashboard
Manage Rooms (CRUD)
View Reservations (see all bookings)
Manage Users (if needed)
Go back to dashboard
Database (MySQL)
hotel_reservation
users
user_id (PK)
name
email
password
role
rooms
room_id (PK)
room_name
price
reservation
reservation_id (PK)
user_id (FK)
room_id (FK)
check_in
check_out
address
contact_number
End

# 4. Conclusion

**In conclusion, the Reservation Module is a key feature of the Online Hotel Reservation System because it provides a faster and more convenient booking process for customers. It minimizes manual work, improves accuracy in reservation management, and enhances the overall experience for everyone using the system.**

**Thank you for listening. We hope this demonstration gives you a clear understanding of how our Online Hotel Reservation System simplifies hotel reservations for all users.**

## Self-Audit

**1. DB operations map clearly to rows?**

- Yes. The system stores customer accounts, room details, and reservation records in their respective database tables. Relationships between users, rooms, and reservations are maintained to ensure accurate data retrieval and updates.

**2. Input filters reject bad entries?**

- Yes. The system validates customer registration details, login credentials, and reservation information before processing transactions. Invalid or incomplete inputs are rejected to maintain data accuracy.

**3. Reports show accurate fields?**

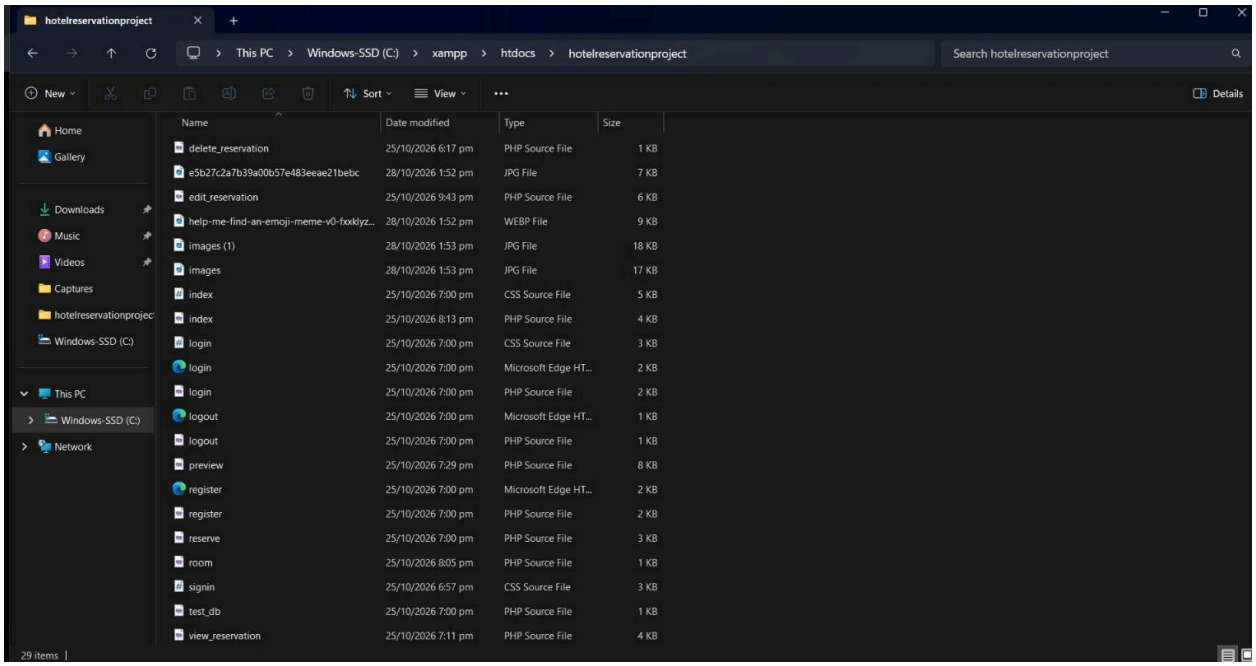
- Yes. The system displays accurate information such as customer details, room descriptions, room prices, check-in and check-out dates, reservSuation status, and total reservation cost based on the stored database records.

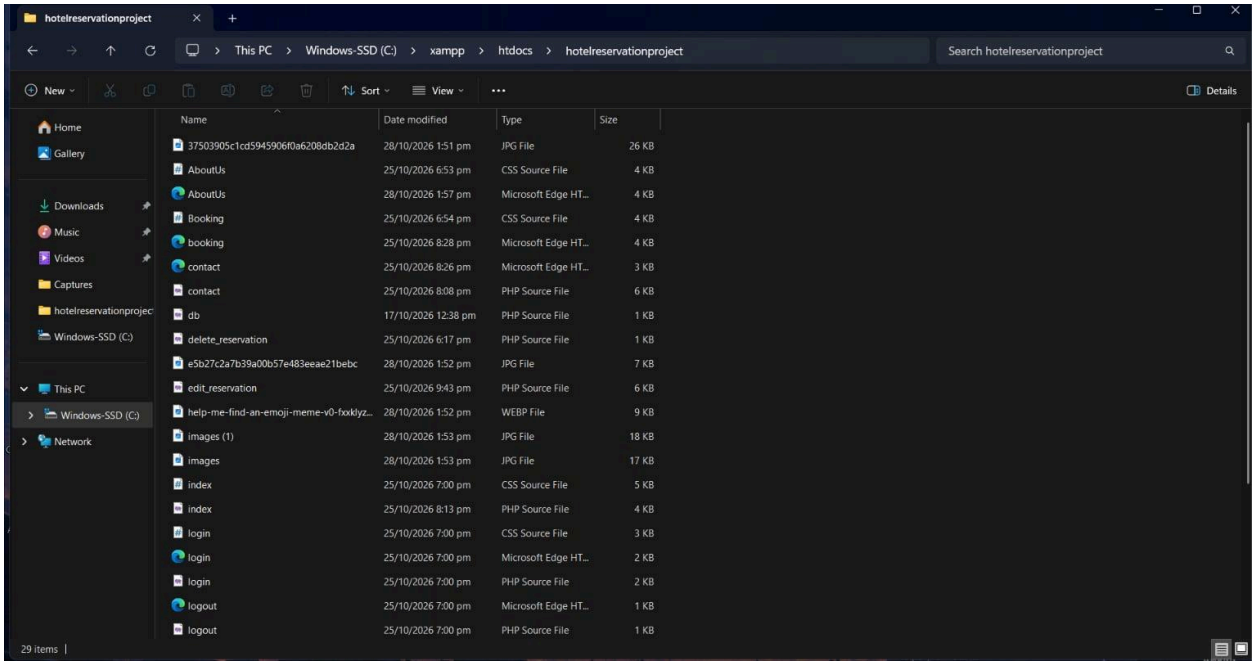
**4. Errors show a clean message?**

- Yes. The system provides clear and understandable messages when login credentials are incorrect, required fields are missing, or a selected room is unavailable, helping users easily identify and correct issues.

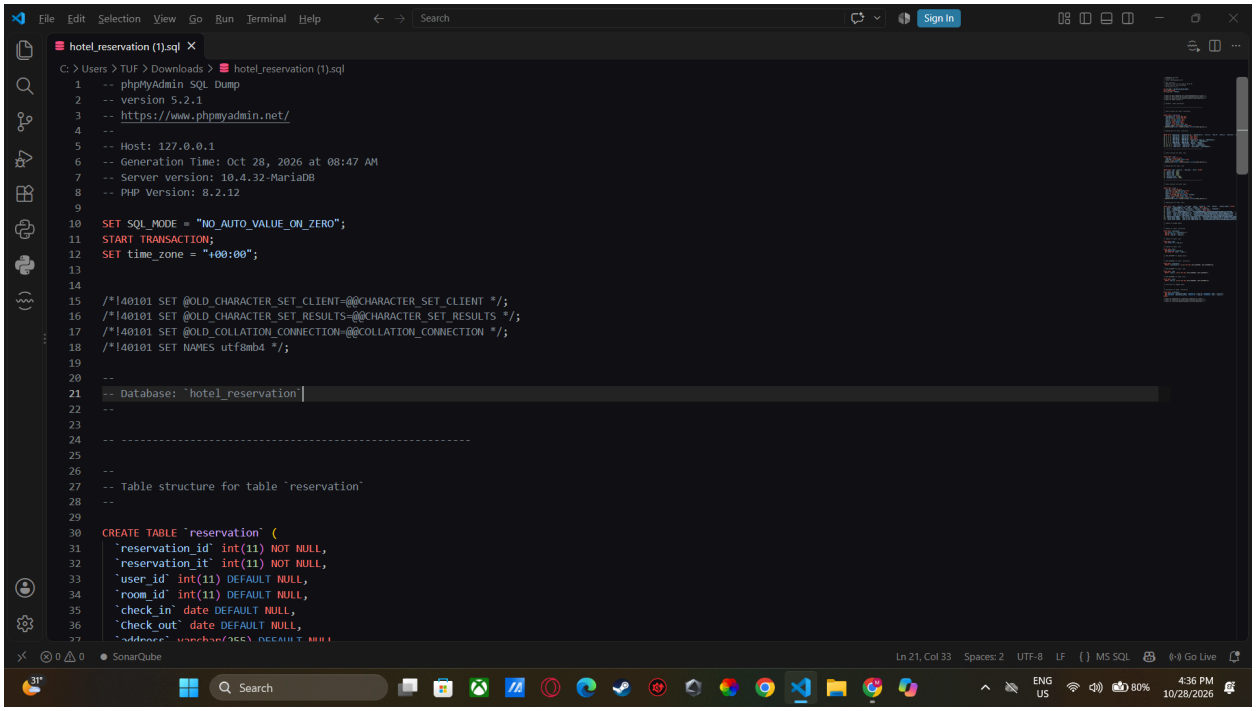
# Submission Package

## 1.Repository Files



# 2.SQL Schema Script



```sql
-- phpMyAdmin SQL Dump
-- version 5.2.1
-- https://www.phpmyadmin.net/
--
-- Host: 127.0.0.1
-- Generation Time: Oct 28, 2026 at 08:47 AM
-- Server version: 10.4.32-MariaDB
-- PHP Version: 8.2.12

SET SQL_MODE = "NO_AUTO_VALUE_ON_ZERO";
START TRANSACTION;
SET time_zone = "+00:00";

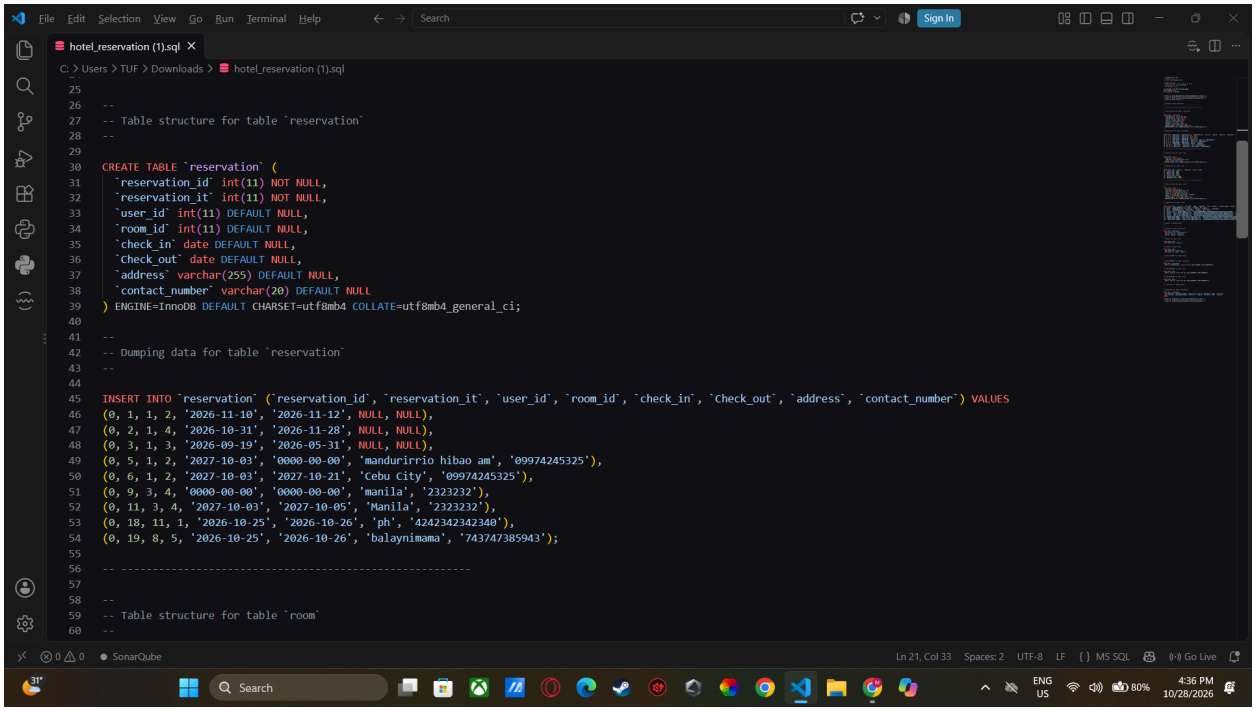

/*!40101 SET @OLD_CHARACTER_SET_CLIENT=@@CHARACTER_SET_CLIENT */;
/*!40101 SET @OLD_CHARACTER_SET_RESULTS=@@CHARACTER_SET_RESULTS */;
/*!40101 SET @OLD_COLLATION_CONNECTION=@@COLLATION_CONNECTION */;
/*!40101 SET NAMES utf8mb4 */;

--
-- Database: `hotel_reservation`
--

-- --------------------------------------------------------

--
-- Table structure for table `reservation`
--

CREATE TABLE `reservation` (
  `reservation_id` int(11) NOT NULL,
  `reservation_it` int(11) NOT NULL,
  `user_id` int(11) DEFAULT NULL,
  `room_id` int(11) DEFAULT NULL,
  `check_in` date DEFAULT NULL,
  `Check_out` date DEFAULT NULL,
```



```sql

--
-- Table structure for table `reservation`
--

CREATE TABLE `reservation` (
  `reservation_id` int(11) NOT NULL,
  `reservation_it` int(11) NOT NULL,
  `user_id` int(11) DEFAULT NULL,
  `room_id` int(11) DEFAULT NULL,
  `check_in` date DEFAULT NULL,
  `Check_out` date DEFAULT NULL,
  `address` varchar(255) DEFAULT NULL,
  `contact_number` varchar(20) DEFAULT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;

--
-- Dumping data for table `reservation`
--

INSERT INTO `reservation` (`reservation_id`, `reservation_it`, `user_id`, `room_id`, `check_in`, `Check_out`, `address`, `contact_number`) VALUES
(0, 1, 1, 2, '2026-11-10', '2026-11-12', NULL, NULL),
(0, 2, 1, 4, '2026-10-31', '2026-11-28', NULL, NULL),
(0, 3, 1, 3, '2026-09-19', '2026-05-31', NULL, NULL),
(0, 5, 1, 2, '2027-10-03', '0000-00-00', 'mandurirrio hibao am', '09974245325'),
(0, 6, 1, 2, '2027-10-03', '2027-10-21', 'Cebu City', '09974245325'),
(0, 9, 3, 4, '0000-00-00', '0000-00-00', 'manila', '2323232'),
(0, 11, 3, 4, '2027-10-03', '2027-10-05', 'Manila', '2323232'),
(0, 18, 11, 1, '2026-10-25', '2026-10-26', 'ph', '4242342342340'),
(0, 19, 8, 5, '2026-10-25', '2026-10-26', 'balaynimama', '743747385943');

-- --------------------------------------------------------

--
-- Table structure for table `room`
--
```

## 3.System Deployment Guide

Sanctovia Hotel Reservation System

System Deployment Guide

Requirements

- XAMPP with PHP and MySQL
- A web browser
- The project folder
- The `hotel_reservation.sql` database file

Installation Steps

1. Install and open XAMPP.
2. Start Apache and MySQL.
3. Open phpMyAdmin at `http://localhost/phpmyadmin/`.
4. Create a database named `hotel_reservation`, if it does not already exist.
5. Import `hotel_reservation.sql` into that database.
6. Copy the `hotelreservationproject` folder into the XAMPP `htdocs` directory.
7. Check that `db.php` uses the correct database name and connection settings.
8. Open `http://localhost/hotelreservationproject/` in a browser.
9. Register an account or log in to test the system.

Basic Testing

- Test user registration and login.
- Test room reservations.
- Test viewing, editing, and deleting reservations.
- Test the logout feature.

Notes

Apache and MySQL must be running while using the system locally. Database credentials and settings may need to be adjusted on another computer.